

GRADED SOLUTIONS:

1.

```
SELECT m.match_num, r.name
FROM matches m
JOIN match_referees mr ON m.match_num = mr.match_num
JOIN referees r ON mr.fourth_referee = r.referee_id
WHERE m.match_date = '2020-05-15'
```
2.

```
SELECT p.name
FROM players p
JOIN teams t ON p.team_id = t.team_id
WHERE t.name = 'All Stars'
ORDER BY p.dob DESC
LIMIT 1
```
3.

```
SELECT team_id
FROM teams
WHERE team_id NOT IN (
    SELECT DISTINCT team_id
    FROM players
    WHERE jersey_no IN (77, 88)
)
```
4.

```
SELECT s.student_fname, bi.member_no
FROM book_issue bi
JOIN members m ON bi.member_no = m.member_no
JOIN students s ON m.roll_no = s.roll_no
WHERE bi.doi = '2021-08-01'
```
5.

```
SELECT bc.title, COUNT(*) AS num_copies
FROM book_catalogue bc
JOIN book_copies bcp ON bc.ISBN_no = bcp.ISBN_no
WHERE bc.title ILIKE '%Easy%'
GROUP BY bc.title
```

PRACTICE SOLUTIONS:

1.

```
SELECT t.name
FROM teams t
INNER JOIN matches m
  ON t.team_id = m.host_team_id
  OR t.team_id = m.guest_team_id
GROUP BY t.team_id, t.name
HAVING COUNT(m.match_num) > 3
```
2.

```
SELECT
DISTINCT faculty_fname, faculty_lname
FROM faculty
NATURAL JOIN members
NATURAL JOIN book_issue
WHERE faculty.department_code = 'ME'
```
3.

```
SELECT
COUNT(b.title)
FROM book_catalogue b
JOIN book_copies c on b.ISBN_no=c.ISBN_no
JOIN book_issue i on i.accession_no=c.accession_no
WHERE i.doi='2021-08-11'
```
4.

```
SELECT faculty_fname,faculty_lname
FROM faculty
EXCEPT
SELECT f.faculty_fname,f.faculty_lname
FROM faculty f
NATURAL JOIN members m
NATURAL JOIN book_issue i
```
5.

```
SELECT
DISTINCT b.title
FROM book_catalogue b inner join
book_copies c on b.ISBN_no=c.ISBN_no
INNER JOIN book_issue i on i.accession_no=c.accession_no
INNER JOIN members m on m.member_no=i.member_no
WHERE member_type='PG'
EXCEPT
```

```
SELECT distinct b.title
FROM book_catalogue b inner join
book_copies c on b.ISBN_no=c.ISBN_no
INNER JOIN book_issue i on i.accession_no=c.accession_no
INNER JOIN members m on m.member_no=i.member_no
WHERE member_type='UG'
```